

UQFF C++ Source Module

Lines: 1,086 Size: 60,987 bytes

Page 1 • Unified Quantum Field Framework (UOFF) v4.81


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190         case SYS_ANYONS_CERN:    return M_any;
191         case SYS_H_RES_FE56:     return M_fe56;
192         case SYS_VAC_HARMONICS:  return M_vac;
193         default:                  return 0.0;
194     }
195 }
196
197 double get_r(int idx) const {
198     switch (idx) {
199         case SYS_W2_TRIADIC:      return r_w2t;
200         case SYS_SGRA_BUOY:       return r_sgra_b;
201         case SYS_PILLARS_DPM:     return r_pil_d;
202         case SYS_RESONANCE_26D:   return r_res26;
203         case SYS_UTE2_SC:         return r_ute2;
204         case SYS_ANYONS_CERN:     return r_any;
205         case SYS_H_RES_FE56:     return r_fe56;
206         case SYS_VAC_HARMONICS:   return r_vac;
207         default:                  return 1.0;
208     }
209 }
210
211 double get_DPM_res(int idx) const {
212     switch (idx) {
213         case SYS_W2_TRIADIC:      return DPM_res_w2;
214         case SYS_SGRA_BUOY:       return DPM_res_sgra;
215         case SYS_PILLARS_DPM:     return DPM_res_pil;
216         default:                  return 1.0;    // other systems: unscaled
217     }
218 }
219
220 /// Effective magnetic field per system (for U_m computation)
221
222 double get_B_eff(int idx) const {
223     switch (idx) {
224         case SYS_W2_TRIADIC:      return 1.0e-5;    // T   typical OB-wind ISM
225         case SYS_SGRA_BUOY:       return 1.0e-3;    // T   Galactic Centre
226         case SYS_PILLARS_DPM:     return 1.0e-5;    // T   ISM pillar
227         case SYS_RESONANCE_26D:   return 1.0e-5;    // T   reference
228         case SYS_UTE2_SC:         return B_ute2;    // T   UTe2 lab field (16 T)
229         case SYS_ANYONS_CERN:     return 1.0;       // T   experiment bench
230         case SYS_H_RES_FE56:     return 30.0;       // T   hyperfine at Fe nucleus
231         case SYS_VAC_HARMONICS:   return 1.0e-9;    // T   IGM background
232         default:                  return 1.0e-5;
233     }
234 }
235
236 const char* system_name(int idx) const {
237     static const char* names[NUM_SYSTEMS_002] = {
238         &"Westerlund 2 [DPM Triadic]&",
239         &"Sgr A* [Buoyancy Dominant]&",
240         &"Pillars of Creation [DPM Resonance]&",
241         &"26D Resonance Layer Framework&",
242         &"UTe2 Topological Superconductor&",
243         &"Anyon Condensate [CERN 2025]&",
244         &"Fe-56 Nucleus [H_res Shell Model]&",
245         &"Cosmic Void [Vacuum Harmonics]&"
246     };

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245     };
246     return (idx >= 0 && idx < NUM_SYSTEMS_002) ? names[idx] : &quot;Unknown&quot;
247 }
248
249 const char* regime_label(int idx) const {
250     static const char* labels[NUM_SYSTEMS_002] = {
251         &quot;OB Star Cluster (6 pc, 30 km\&times;x98\&times;x89)&quot;;,
252         &quot;SMBH (1.27\&times;x9710\&times;x2\&times;b9\&times;x2\&times;b0 m, 4\&times;x9710\&times;x2\&times;x81\&times;b6 M\&times;x98\&times;x89)&quot;;,
253         &quot;Star-Forming Pillar (5 pc, ~10\&times;x2\&times;x81\&times;b4 M\&times;x98\&times;x89)&quot;;,
254         &quot;26 Orthogonal Vacuum States (Solar ref.)&quot;;,
255         &quot;Condensed Matter – 1 nm crystal lattice&quot;;,
256         &quot;Quasiparticle – fm scale (nuclear)&quot;;,
257         &quot;Atomic Nucleus – 4.35 fm (Fe-56)&quot;;,
258         &quot;Cosmic Void – 1 Mpc scale&quot;;
259     };
260     return (idx >= 0 && idx < NUM_SYSTEMS_002) ? labels[idx] : &quot;Unknown&quot;
261 }
262
263 const char* unique_physics_str(int idx) const {
264     static const char* phys[NUM_SYSTEMS_002] = {
265         &quot;DPM integral F_U_Bi_i = int[LENR+DPM_mom+DPM_grav+...] dx ~2.11e208 N; &quot;;
266         &quot;DPM_res=1.67e3; F_LEN=1.56e36 N; x2=-1.35e172 m (quadratic); T_sf=2 Myr.&quot;;,
267
268         &quot;F_U_Bi_i~-8.31e211 N (F_LEN dominant); DPM_res=1.67e7; NSC tidal; &quot;;
269         &quot;Kerr spin correction a*=0.3; QPO 20-min; GW back-reaction dOmega/dt.&quot;;,
270
271         &quot;DPM_res=1.67e7 (UV-enhanced); F_U_Bi_i~2.11e212 N; triadic F_Ubi=9.79e-33 N; &quot;;
272         &quot;f_z,CGM~1.46e-73 ([SSq]-corrected); JWST NIRC&times; 2022 pillar density.&quot;;,
273
274         &quot;R(t)=sum_{i=1}^{26} R_i*cos(omega_i*t); R_i=F_base*(1+SSq)*exp(-SSq*i/26); &quot;;
275         &quot;omega_i=2pi/T_sf*i*(1+SSq); [SSq]=0.57; 26 orthogonal vacuum quantum states.&quot;;
276
277         &quot;delta_{n,UTe2}=(2pi)^{n/6}*exp(-SSq*n/26)*(1+f_topo)*exp(-pi); &quot;;
278         &quot;B_threshold=16 T; f_topo=0.20; Andreev STM; non-unitary TQFT; n=1..9.&quot;;,
279
280         &quot;F_UBii,anyons=-F_rel*(E_any/E_LEP)*Q_wave*g*exp(-delta_c^2/(2*sigma^2)); &quot;;
281         &quot;delta_c=1.686; sigma=1.0; Ising braiding E_any=1 MeV; 75% CERN 2025 alignment.&quot;;
282
283         &quot;H_res=A_res*sin(2pi*f_res*t)+U_dp*SC_m*k_nuc+S_shell; Z=26,A=56,N=30; &quot;;
284         &quot;S_shell=0.20 (near magic 28); Fe-56 binding 8.79 MeV/nucleon (global peak).&quot;;
285
286         &quot;U_g2=sum H_m*(1-exp(-SSq*m))*cos(omega*t_n); H_m=sum(1/k)*f_Ub; &quot;;
287         &quot;V_series=sum(1/n^26)*SSq^n~0.570=Li26(SSq); dynamic SSq(n,t).&quot;;
288     };
289     return (idx >= 0 && idx < NUM_SYSTEMS_002) ? phys[idx] : &quot;&quot;;
290 }
291
292 void get_ext(int idx, double& M_ext, double& r_ext) const {
293     switch (idx) {
294         case SYS_W2_TRIADIC: M_ext = M_ext_w2t; r_ext = r_ext_w2t; break;
295         case SYS_SGRA_BUOY: M_ext = M_ext_sgra_b; r_ext = r_ext_sgra_b; break;
296         case SYS_PILLARS_DPM: M_ext = M_ext_pil_d; r_ext = r_ext_pil_d; break;
297         case SYS_RESONANCE_26D: M_ext = M_ext_res26; r_ext = r_ext_res26; break;
298         case SYS_UTE2_SC: M_ext = M_ext_ute2; r_ext = r_ext_ute2; break;
299         case SYS_ANYONS_CERN: M_ext = M_ext_any; r_ext = r_ext_any; break;

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300         case SYS_H_RES_FE56:      M_ext = M_ext_fe56;    r_ext = r_ext_fe56;    break;
301         case SYS_VAC_HARMONICS: M_ext = M_ext_vac;      r_ext = r_ext_vac;    break;
302         default: M_ext = M_sun; r_ext = 1.0e11; break;
303     }
304 }
305
306 // =====
307 // Initialise all system default parameters
308 // =====
309 void initializeDefaults() {
310     // ■ [0] Westerlund 2 – DPM triadic ■=====
311     M_w2t      = 30000.0 * M_sun; // 3.0e34 kg
312     r_w2t      = 1.89e16;          // m (6 pc cluster radius)
313     DPM_res_w2 = 1.67e3;           // dimensionless
314     F_LEN_R_w2 = 1.56e36;          // N
315     T_sf_w2    = 6.307e13;         // s (2 Myr)
316     M_ext_w2t  = 4.0e6 * M_sun;    // SgrA* tidal (8 kpc distance)
317     r_ext_w2t  = 2.469e20;         // m (~8 kpc)
318
319     // ■ [1] Sgr A* – buoyancy dominant ■=====
320     M_sgra_b    = 4.0e6 * M_sun;    // 7.956e36 kg
321     r_sgra_b    = 1.27e10;          // m (~0.09 AU, G2 cloud periastron)
322     DPM_res_sgra = 1.67e7;          // NSC-enhanced
323     F_LEN_R_sgra = 6.16e39;         // N (F_LEN_R >> F_rel at this scale)
324     T_sf_sgra    = 3.156e11;        // s (10 kyr accretion cycle)
325     M_ext_sgra_b = 1.0e11 * M_sun;  // NSC mass
326     r_ext_sgra_b = 9.461e16;        // m (~3 pc, NSC half-light radius)
327
328     // ■ [2] Pillars of Creation – DPM resonance ■=====
329     M_pil_d     = 10100.0 * M_sun;  // 2.009e34 kg
330     r_pil_d     = 4.73e16;          // m (5 pc pillar height)
331
332     DPM_res_pil = 1.67e7;           // UV-ionisation enhanced
333     F_LEN_R_pil = 1.56e36;          // N
334     T_sf_pil    = 3.156e13;         // s (1 Myr)
335     M_ext_pil_d = 40.0 * M_sun;     // Eta Carinae (η Car) + Trumpler 14
336     r_ext_pil_d = 6.17e16;          // m (~6.5 pc separation)
337
338     // ■ [3] 26D resonance layer ■=====
339     M_res26     = 1.0 * M_sun;      // reference: 1 M☉
340     r_res26     = 6.957e8;          // m (R_sun – reference scale)
341     N_layers_res26 = 26.0;          // 26 quantum vacuum states
342     T_sf_res26    = 3.156e13;       // s (1 Myr – resonance period)
343     SSq_res26     = SSq;            // 0.57 (calibrated [SSq])
344     M_ext_res26   = 4.0e6 * M_sun;  // SgrA* (cosmological boundary)
345     r_ext_res26   = 1.27e10;        // m (ISCO scale)
346
347     // ■ [4] UTe2 topological superconductor ■=====
348     M_ute2      = m_e_kg;           // 9.109e-31 kg (electron mass as reference)
349     r_ute2      = 1.0e-9;           // m (1 nm crystal unit cell)
350     B_ute2      = 16.0;             // T (superconducting threshold)
351     f_topo_ute2 = 0.20;             // dimensionless (centre 0.1-0.3 range)
352     M_ext_ute2  = m_p_kg;           // proton mass (nuclear context)
353     r_ext_ute2  = 1.0e-10;          // m (Bohr radius, atomic scale)
354
355     // ■ [5] Anyon condensate (CERN 2025) ■=====

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740     });
741
742     for (int i : order) {
743         double g    = compute_g_base(i);
744         double ubi   = compute_Ubi(i);
745         double fii   = compute_F_UBii(i, t);
746         double um    = compute_Um_full(i, t);
747         std::cout << << std::left
748             << << std::setw(4) << << i
749             << << std::setw(34) << << system_name(i)
750             << << std::setw(16) << << std::scientific << << std::setprecision(3)
751             << << std::setw(16) << << ubi
752             << << std::setw(18) << << fii
753             << << std::setw(18) << << um << << << "\n";
754     }
755     std::cout << << << "\n";
756     std::cout << << << " NOTE: F_UBii shown is Tier-2 standard (DPM-scaled for sys[0-2])\n";
757     std::cout << << << " For calibrated DPM integral use compute_DPM_log_scale(idx).\n";
758     std::cout << << << " For anyon: compute_F_UBii_anyons(); 26D: compute_26D_resonance.\n";
759 }
760
761 // Print compact parameter dump for all 8 systems.
762 void printParameters() const {
763     std::cout << << << "\n--- UQFFLearningAssessment002 Parameters ---\n";
764     std::cout << << std::scientific << << std::setprecision(4);
765     auto row = [&](const char* tag, double v) {
766         std::cout << << << " << << std::left << << std::setw(22) << << tag << << v << << "\n";
767     };
768     std::cout << << << "[0] Westerlund 2 Triadic\n";
769     row("<M_w2t<", M_w2t); row("<r_w2t<", r_w2t);
770     row("<DPM_res_w2<", DPM_res_w2); row("<F_LEN_R_w2<", F_LEN_R_w2);
771     row("<T_sf_w2<", T_sf_w2);
772     std::cout << << << "[1] Sgr A* Buoyancy\n";
773     row("<M_sgra_b<", M_sgra_b); row("<r_sgra_b<", r_sgra_b);
774     row("<DPM_res_sgra<", DPM_res_sgra); row("<F_LEN_R_sgra<", F_LEN_R_sgra);
775     std::cout << << << "[2] Pillars DPM\n";
776     row("<M_pil_d<", M_pil_d); row("<r_pil_d<", r_pil_d);
777     row("<DPM_res_pil<", DPM_res_pil); row("<F_LEN_R_pil<", F_LEN_R_pil);
778     std::cout << << << "[3] 26D Resonance\n";
779     row("<M_res26<", M_res26); row("<r_res26<", r_res26);
780     row("<N_layers<", N_layers_res26); row("<T_sf_res26<", T_sf_res26);
781     row("<SSq_res26<", SSq_res26);
782     std::cout << << << "[4] UTe2 SC\n";
783     row("<M_ute2<", M_ute2); row("<r_ute2<", r_ute2);
784     row("<B_ute2<", B_ute2); row("<f_topo_ute2<", f_topo_ute2);
785     std::cout << << << "[5] Anyons CERN\n";
786     row("<M_any<", M_any); row("<r_any<", r_any);
787     row("<E_anyons_loc<", E_anyons_loc); row("<delta_c_loc<", delta_c_loc);
788     std::cout << << << "[6] H_res Fe-56\n";
789     row("<M_fe56<", M_fe56); row("<r_fe56<", r_fe56);
790     row("<Z_fe56<", Z_fe56); row("<A_fe56<", A_fe56);
791     row("<N_fe56<", N_fe56);
792     std::cout << << << "[7] Vacuum Harmonics\n";
793     row("<M_vac<", M_vac); row("<r_vac<", r_vac);
794     row("<N_harmonics<", N_harmonics_vac); row("<gamma_vac<", gamma_vac_loc);

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1015         auto it = dynamic_params.find(key);
1016         if (it != dynamic_params.end()) return it->second;
1017         if (logging_enabled)
1018             std::cout <<< <<< "[LOG] Dynamic param not found: " <<< key <<< <<< "\n";
1019         return std::numeric_limits<double>::quiet_NaN();
1020     }
1021
1022     void exportState(const std::string& filename = "UQFFLearningAssessment002_state.txt&");
1023     std::ofstream ofs(filename);
1024     if (!ofs) {
1025         std::cerr <<< "Error: Cannot open " <<< filename <<< "\n";
1026         return;
1027     }
1028     ofs <<< "# UQFFLearningAssessment002 (PhD Research Edition) state export\n";
1029     ofs <<< std::scientific <<< std::setprecision(10);
1030     const char* names[] = {
1031         // [0] W2
1032         "M_w2", "r_w2", "DPM_res_w2", "F_LEN_R_w2", "M_ext_w2", "r_ext_w2",
1033         // [1] SgrA*
1034         "M_sgra_b", "r_sgra_b", "DPM_res_sgra", "F_LEN_R_sgra", "M_ext_sgra_b", "r_ext_sgra_b",
1035         // [2] Pillars
1036         "M_pil_d", "r_pil_d", "DPM_res_pil", "F_LEN_R_pil", "M_ext_pil_d", "r_ext_pil_d",
1037         // [3] 26D
1038         "M_res26", "r_res26", "N_layers_res26", "T_sf_res26", "M_ext_res26", "r_ext_res26",
1039         // [4] UTe2
1040         "M_ute2", "r_ute2", "B_ute2", "f_topo_ute2", "M_ext_ute2", "r_ext_ute2",
1041         // [5] Anyons
1042         "M_any", "r_any", "E_anyons_loc", "delta_c_loc", "M_ext_any", "r_ext_any",
1043         // [6] Fe-56
1044         "M_fe56", "r_fe56", "Z_fe56", "A_fe56", "M_ext_fe56", "r_ext_fe56",
1045         // [7] Vac
1046         "M_vac", "r_vac", "N_harmonics_vac", "gamma_vac_loc", "M_ext_vac", "r_ext_vac",
1047         // metrics
1048         "diversity_score", "dynamic_score", "scalability_score",
1049     };
1050     for (const char* n : names)
1051         ofs <<< n <<< " = " <<< getVariable(n) <<< "\n";
1052     for (const auto& kv : dynamic_params)
1053         ofs <<< "dynamic." <<< kv.first <<< " = " <<< kv.second <<< "\n";
1054     if (logging_enabled)
1055         std::cout <<< "[LOG] State exported to: " <<< filename <<< "\n";
1056 }
1057
1058 /// Cross-validate g_base[sys_idx] against an external UQFF module.
1059 /// Store the other module's g beforehand: setDynamicParameter("cross_g", val).
1060 template<typename OtherModuleT>
1061 double cross_validate(const OtherModuleT& other, int sys_idx,

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```

1070             double /*t_seconds*/ = 0.0) const {
1071     double g_here = compute_g_base(sys_idx);
1072     double g_other = 0.0;
1073     auto it = dynamic_params.find("cross_g");
1074     if (it != dynamic_params.end()) g_other = it->second;
1075     double frac = (g_here != 0.0)
1076         ? std::abs(g_other - g_here) / std::abs(g_here)
1077         : std::numeric_limits<double>::quiet_NaN();
1078     if (logging_enabled)
1079         std::cout << " [XVAL002] sys[" << sys_idx << " ] g_here="
1080             << g_here << " g_other=" << g_other
1081             << " frac_diff=" << frac << " \n";
1082     return frac;
1083 }
1084 };
1085
1086 #endif // UQFF_LEARNING_ASSESSMENT_002_H

```

— End of UQFFLearningAssessment_002.cpp —